Sequence Diagram Specification

for

VisionEdge: AI-Powered Optical Store Management & Virtual Try-On System

Version 1.0

File Created: visionedge_sequence_diagram.drawio

Theme: White Background, Black Typography (#000000)

Date: August 2026

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Version: 1.0

File Created: [visionedge_sequence_diagram.drawio](file:///c:/projects/VisionEdge/visionedge_sequence_diagram.drawio)

Format: Draw.io (XML Schema)

Theme: Crisp White Background, High-Contrast Black Typography (fontColor=#000000), Soft Indigo Lifelines (#4F46E5 / #EEF2FF)

Date: August 2026

1. Sequence Diagram Visual Representation

The sequence diagram documents the complete end-to-end operational flow of VisionEdge, tracing customer and optometrist interactions through the Web System Interface, Python Flask API Controller, Gemini 1.5 Pro AI Vision Engine, MediaPipe Virtual Try-On Suite, MySQL Relational Database, and POS Billing Engine:

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sequenceDiagram
autonumber
actor Customer as Customer / Staff
participant UI as System Interface (Web UI)
participant Ctrl as API Controller (Flask Backend)
participant AI as AI & CV Engine (Gemini / MediaPipe)
participant DB as Database (MySQL / SQLAlchemy)
participant POS as POS & Invoice Engine (PDF / Tax)

Customer->>UI: Request available eye test slots & catalog
UI->>Ctrl: Fetch appointments & frame list (GET /api/appointments)
Ctrl->>DB: Query doctor schedules & frame inventory
DB-->>Ctrl: Return slot availability & frame stock records
Ctrl-->>UI: Send schedules & frame catalog JSON
UI-->>Customer: Display appointment calendar & frame options

Customer->>UI: Upload physical prescription scan image
UI->>Ctrl: Initiate AI extraction (POST /api/ai/extract)
Ctrl->>AI: Transmit image payload to Google Gemini 1.5 Pro AI
AI-->>Ctrl: Return parsed diopters JSON (OD/OS SPH, CYL, AXIS, PD, Confidence)
Ctrl->>DB: Save EyeTest & Prescription records
DB-->>Ctrl: Confirm record creation (Prescription ID)
Ctrl-->>UI: Return auto-filled prescription & lens recommendations

Customer->>UI: Launch 3D Virtual Try-On & select frame
UI->>AI: Detect facial landmarks via MediaPipe & position frame PNG
AI-->>UI: Render live 3D frame overlay on webcam canvas

Customer->>UI: Confirm frame & lens selection (Proceed to POS Checkout)
UI->>Ctrl: Initiate billing transaction (POST /api/billing/create-invoice)
Ctrl->>POS: Calculate 18% GST tax, discounts & compile PDF invoice
POS->>DB: Deduct frame/lens inventory stock & store invoice record
```

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DB-->>POS: Invoice transaction committed (Invoice ID)
POS-->>Ctrl: Return PDF invoice URL & payment status
Ctrl-->>UI: Send order completion payload
UI-->>Customer: Display order success message & PDF invoice download button
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2. Sequence Interaction Steps Matrix

Step #	Origin	Destination	Action / API Endpoint	Payload / Data Returned
1	Customer / Staff	System Interface	Request Eye Test Slots & Catalog	User Click / Page Request
2	System Interface	API Controller	GET /api/appointments/slots, GET /api/inventory/frames	HTTP GET Request
3	API Controller	Database	SELECT * FROM appointment, SELECT * FROM frame	SQL Query Execution
4	Database	API Controller	Return Record Sets	SQL Result Array
5	API Controller	System Interface	HTTP 200 OK	Schedule & Frame Inventory JSON
6	System Interface	Customer / Staff	Render Calendar & Catalog View	Dynamic HTML UI Display
7	Customer / Staff	System Interface	Upload Prescription Image Scan	File Selection (JPG/PNG)
8	System Interface	API Controller	POST /api/ai/extract	Multipart Form Data
9	API Controller	Gemini AI Engine	Google Generative AI Multimodal Request	Image Buffer + JSON Prompt Schema
10	Gemini AI Engine	API Controller	Optical Diopters JSON	SPH, CYL, AXIS, ADD, PD, Confidence (98%)
11	API Controller	Database	INSERT INTO eye_test, INSERT INTO prescription	SQL Insert Transaction
12	Database	API Controller	Record Insertion Confirmation	Generated prescription_id
13	API Controller	System Interface	HTTP 201 Created	Extracted Diopters & Lens Recommendations
14	Customer / Staff	System Interface	Select Frame & Launch Virtual Try-On	Canvas Frame Selection
15	System Interface	MediaPipe Engine	FaceMesh.send({ image: videoElement })	Live Camera Video Frame Stream
16	MediaPipe Engine	System Interface	Facial Mesh Landmarks Grid	Real-time 2D/3D Canvas Frame Overlay
17	Customer / Staff	System Interface	Confirm Frame & Proceed to Checkout	Form Submit
18	System Interface	API Controller	POST /api/billing/create-invoice	Order Lines (Frame ID, Lens ID, Customer ID)
19	API Controller	POS Invoice Engine	Calculate Tax & Generate Invoice PDF	Subtotal, Discount, 18% GST, FPDF Compilation
20	POS Invoice Engine	Database	UPDATE frame SET stock = stock - 1, INSERT INTO billing	SQL Stock Reduction & Billing Commit
21	Database	POS Invoice Engine	Transaction Committed	Database Commit Acknowledgment
22	POS Invoice Engine	API Controller	Generated Invoice PDF File	File Path invoice_INV_2026_001.pdf
23	API Controller	System Interface	HTTP 200 OK	Order Status & PDF Download URL
24	System Interface	Customer / Staff	Display Success Screen & Printable Receipt	Order Confirmation Screen

3. How to Use & Open in Draw.io

1. Open your browser and go to app.diagrams.net (Draw.io).
2. Click File -> Open From -> Device / Local File.
3. Select the file: c:\projects\VisionEdge\visionedge_sequence_diagram.drawio.
4. You can customize, export to PNG/SVG/PDF, or edit lifelines directly.

End of VisionEdge Sequence Diagram Specification.